

Application-Specific Flexible PCB Sensor Integration for Multi-Modal Feedback in Soft Robotic Grippers

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Abstract—Soft robotic grippers are remarkably good at handling objects that traditional rigid robots struggle with - things that are fragile, irregularly shaped, or difficult to predict. The problem is that making them smart is harder than making them flexible. Conventional sensors, circuit boards, and wiring are built for rigid systems, and forcing them into a soft gripper often creates the very stiffness the design was meant to avoid. External cameras offer another option, however they lose sight of the action the moment the gripper closes around an object. The proposed architecture addresses this by building a flexible printed circuit board (FPCB) around the physical realities of how a soft finger bends and makes contact. A gripper-shaped polyimide board and a shared I2C bus support motion, pressure, and temperature sensing with minimal wiring. The design is a prototype sensing architecture intended for future fabrication, elastomer integration, and closed-loop grasp validation.

Index Terms—Flexible printed circuit board, FPCB, soft robotics, soft gripper, multi-modal sensing, embedded sensing, I2C, grasp feedback.

I. INTRODUCTION

SOFT robotic grippers work by conforming to objects rather than forcing them into a fixed grasp, making them well-suited for handling things that are fragile, irregular, or unpredictable [1]. The challenge is that the hardware needed to make them responsive has to bend with the gripper, avoid creating stiff spots, and still deliver meaningful feedback.

Most conventional sensing approaches work against these requirements. Rigid sensor modules introduce local stiffness inside a finger that is supposed to flex freely. Extra wiring resists bending and tends to fail over repeated use. External cameras seem like a reasonable workaround until the gripper or object blocks the line of sight, at which point recovering useful information requires additional computation that brings its own complications [3].

The architecture proposed here addresses this through application-specific integration rather off-the-shelf adaptation. Flexible electronics and embedded sensing are not new to soft robotics - the distinction lies in designing the board geometry, sensor placement, and shared-bus communication specifically around the physical behavior of a soft gripper, so the sensing layer works with the structure rather than in spite of it.

The main contributions of this prototype report are: (1) a gripper-oriented FPCB topology for soft robotic integration; (2) a multi-modal sensing layout combining motion, pressure-proxy, and temperature feedback; (3) a reduced-wiring I2C

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This work presents a prototype design concept for flexible PCB-based embedded sensing in soft robotic grippers within the ELIXR Lab research context at Toronto Metropolitan University.

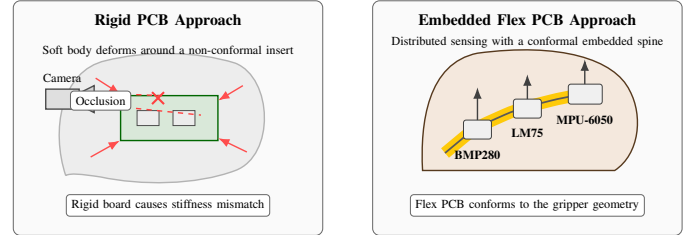


Fig. 1. Comparison between a rigid embedded PCB approach and the proposed flexible PCB approach for soft robotic grippers. A rigid board can create stiffness mismatch and camera-dependent sensing limitations, whereas a conformal flex PCB can wrap with the gripper geometry while supporting embedded multi-modal sensing.

strategy for distributed sensor nodes; and (4) a validation roadmap for future fabrication, embedding, and grasp testing.

II. DESIGN MOTIVATION AND NOVELTY

The motivation for the proposed architecture is summarized in Fig. 1. A soft gripper needs feedback from inside the deforming structure, but conventional sensing can work against the gripper mechanics. A rigid board can create a stiff insert, wiring can limit motion, and a camera can lose sight of the contact area during grasping.

The proposed design responds to these issues by treating the FPCB as part of the gripper structure, not just as a place to mount electronics. The board shape, sensor locations, and I2C routing are chosen together so the sensing system can follow the soft finger instead of forcing a generic rectangular PCB into it.

III. APPLICATION-SPECIFIC FPCB ARCHITECTURE

The proposed FPCB is a narrow sensing spine that can follow the internal profile of a soft gripper finger. As shown in Fig. 2, the layout carries power and communication along the gripper length while placing each sensor where its measurement is expected to be most useful.

The distal region is optimal for inertial sensing - motion and bending are most pronounced near the moving end of the finger, making it the natural place to capture that data. The central region offers a practical location for temperature monitoring, while the pressure sensor serves a local pressure-proxy node once the final embedding geometry is established. Together, these placements make the FPCB a purpose-built sensing layer rather than something bolted on after the fact.

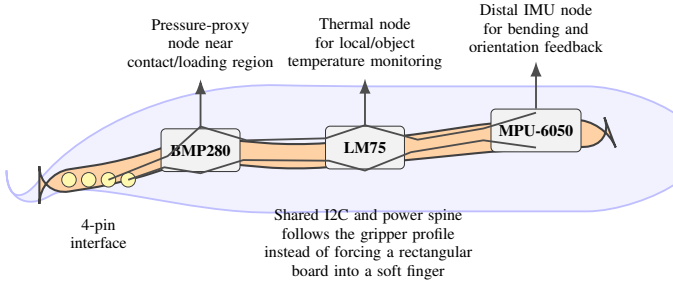


Fig. 2. Application-specific FPCB topology. The prototype board is arranged as a sensing spine, with sensor placement selected according to the local physical quantity being measured rather than arbitrary component placement on a generic PCB.

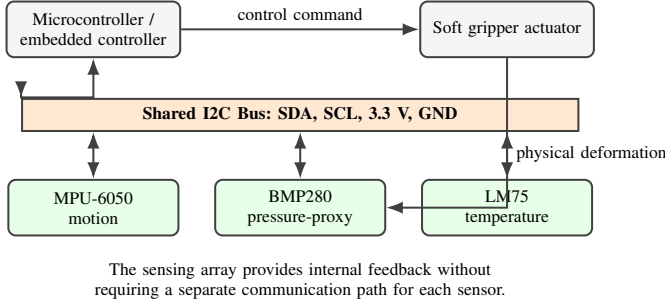


Fig. 3. Multi-modal sensing architecture. Motion, pressure-proxy, and temperature feedback are combined on a shared I2C bus so the soft gripper can be monitored through internal embedded signals rather than relying only on external observation.

IV. MULTI-MODAL EMBEDDED SENSING

A single sensing mode does not fully describe the state of a soft gripper. Motion data can show bending, but not contact pressure or temperature. Pressure data can show local loading, but not global finger orientation. The proposed design therefore combines three complementary sensing modes:

- **Motion feedback:** The MPU-6050 captures acceleration and angular-rate information that can help estimate gripper bending and motion.
- **Pressure-proxy feedback:** The BMP280 provides local pressure information that may be correlated with pneumatic loading, deformation, or contact after calibration.
- **Thermal feedback:** The LM75 provides local temperature monitoring for object interaction, environment awareness, or material protection.

The system-level flow is shown in Fig. 3. The controller polls the distributed sensors over I2C and can use the combined feedback for monitoring or future closed-loop grasp control. Prior work has shown the value of inertial signals for predicting failures in soft hands [2]; this prototype extends that direction by combining inertial feedback with pressure-proxy and thermal sensing on one flexible platform.

V. FLEXIBLE SUBSTRATE AND EMBEDDED INTEGRATION

The substrate choice matters because the electronics should not fight the gripper motion. A rigid FR4 board can create a stiff spot inside an elastomeric structure. A thin polyimide

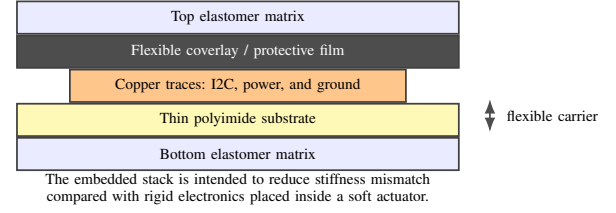


Fig. 4. Conceptual embedded FPCB stack-up. The thin polyimide substrate and protected copper trace layer form a flexible electronic spine within the surrounding elastomeric gripper material.

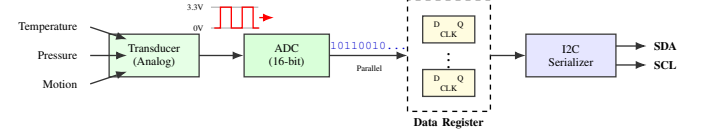


Fig. 5. RTL-level abstraction of the embedded sensing data path. Physical quantities are transduced, digitized, stored in internal data registers, and serialized over the I2C interface for controller access.

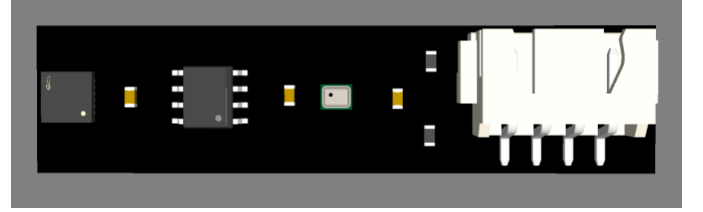


Fig. 6. 3D render of the proposed FPCB sensor prototype. The design uses an elongated flexible form factor to support distributed sensing along a soft robotic gripper.

FPCB is more suitable because it can bend with the surrounding material. Work in flexible and stretchable electronics identifies substrate mechanics, bending strain, and interconnect design as key issues when electronics are placed on non-rigid systems [4], [5].

The proposed stack-up is shown in Fig. 4. The coverlay protects the copper traces, while the polyimide layer acts as the flexible mechanical carrier. Future versions may require serpentine traces or strain-relief regions near high-bending areas, especially around component pads and connector points.

VI. RTL ABSTRACTION OF SENSOR DATA FLOW

Although the prototype uses commercial digital sensors, the RTL-level abstraction in Fig. 5 helps explain the signal path. A physical quantity is sensed, converted into digital data, stored in internal registers, and sent to the controller over I2C. This figure is an abstraction of the data flow, not a claim that a custom ASIC was designed.

VII. PROTOTYPE RENDER AND DESIGN INTERPRETATION

Fig. 6 shows the current 3D render of the proposed FPCB prototype. The render helps communicate the physical design intent: this is not a generic sensor board, but an elongated board form factor intended to support distributed sensing along a soft gripper.

The current prototype should be interpreted as a design-stage platform. Its main value is the integration strategy: a

flexible substrate, shared bus architecture, and application-specific placement of multiple sensing modes. The next step is to fabricate the FPCB and test whether the embedded design preserves gripper compliance while producing useful sensor data.

VIII. EXPECTED ADVANTAGES, LIMITATIONS, AND VALIDATION ROADMAP

The expected advantage of the proposed architecture is that sensing can be added without treating the soft gripper like a rigid mechanical frame. The flexible substrate can reduce stiffness mismatch, the shared I2C bus can reduce wiring, and the internal sensors can keep providing feedback even when a camera view is blocked. Multi-modal data also gives a richer view of gripper state than a single sensing mode.

The main limitation is that this is still a prototype design. It has not yet been fabricated or tested inside a soft gripper. The BMP280 should be treated as a pressure-proxy sensor until its response is calibrated after embedding. The durability of traces, solder joints, and component packages under repeated bending also needs to be tested.

Future validation should proceed in four stages. First, the fabricated FPCB should be tested under repeated bending to check trace and solder-joint durability. Second, the gripper should be tested with and without the FPCB to measure any change in compliance. Third, the MPU-6050, BMP280, and LM75 readings should be calibrated after embedding. Finally, grasp trials should compare whether multi-modal feedback improves grasp-state estimation compared with a single sensing mode or external observation alone.

IX. SOCIETAL AND APPLICATION RELEVANCE

Embedded sensing in soft grippers can support applications where rigid manipulation is unsafe or unreliable. In industrial settings, compliant grippers can handle irregular objects while reducing the chance of damage. In hazardous environments such as mining or remote inspection, robotic manipulation can reduce direct human exposure, although automation also raises labour and economic concerns [6].

The same design direction can also apply to healthcare and assistive robotics. Prosthetic or assistive hands benefit from sensing that is close to the contact surface, especially when manipulating fragile objects or interacting with people. The broader idea is to build the electronics around the soft structure instead of forcing the soft structure to accommodate rigid electronics.

X. CONCLUSION

This report presented a prototype FPCB architecture for multi-modal sensing in soft robotic grippers. The design focuses on a practical integration problem: adding feedback without creating stiff spots, excessive wiring, or dependence on camera visibility. By combining an application-specific board shape, shared I2C communication, and distributed motion, pressure-proxy, and temperature sensors, the prototype provides a clear path toward embedded feedback for future soft gripper control. The work is not yet experimentally validated,

so the next step is fabrication, embedding, calibration, and grasp testing.

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